



Identity, Reset, and Input

Team studio sheet - switch Driver and Navigator every challenge.

Team members: _____

Date: _____

Our numbered kit and device

Kit ____ Device ____ Sensor/type ____ Port ____ Driver ____ Navigator ____

Challenge 1 - prove identity and safe reset

Predict	Run	Evidence	Safe reset
Which prepared output will respond?	Trigger the mentor-approved test.	Write or sketch what proved this is our robot.	Output off / controller ready / roles switch

Challenge 2 - measure an input before choosing a threshold

Condition	Prediction	Actual sensor value	Same unit?	What we noticed
Condition A			Yes / fix	
Condition B			Yes / fix	
A again			Yes / fix	

Our threshold and why

Define A/B conditions and units: _____. Threshold ____; comparison < / > because _____.

Team checkpoint

Both teammates can point to the real readings and explain why copying another team's threshold may not work.



Outputs and a Full Interaction

Use tested inputs/outputs. Two LEDs are the fallback when movement is deferred.

Team members: _____

Date: _____

Challenge 3 - two outputs; record their actual types

Output	Prediction	What we ran	What actually happened	Safe state
Output 1		LED port ____; 0% -> 50% -> 0%		0%
Output 2		Servo HOME/ACTIVE, or second LED 0% -> 50% -> 0%		HOME / LED 0%
Together		Predict, then run both		Both reset

Challenge 4 - input -> decision -> two outputs -> reset

Say the rule before running it

IF input ____ meets ____, THEN output 1 ____ AND output 2 ____; OTHERWISE ____ . Motion: tested / deferred.

Trial	Input condition	Prediction	Output 1	Output 2	Did reset work?
1	Condition A				
2	Condition B				
3	A again				

Change one approved variable, then retest

We changed only ____ . We held ____ the same. The result was...

Final proof

Both explain input, decision, outputs, and reset. Two LEDs are one output type. Label any untested sensor/motion as deferred.



Three Interactions Worth Discussing

Each idea needs an input, a decision, an output, and a reason people would care.

Team members: _____

Date: _____

Problem, delight, or future possibility

Who is the interaction for, and what should be easier, safer, clearer, or more interesting?

Sketch three different IF / THEN systems

1

It responds when...

2

It responds when...

3

It responds when...

Idea	Input	Decision / threshold	Output(s)	Why it matters
1				
2				
3				

Choose or combine

We want to explore idea(s) _____ because the interaction would...



Map the System and First Test

Build the smallest safe interaction that can answer one useful question.

Team members: _____

Date: _____

INPUT	DECISION	OUTPUT 1	OUTPUT 2	OTHERWISE / RESET
What changes in the world?	What condition or threshold?	What happens?	What else happens?	How does it return safely?

Three-frame interaction storyboard

BEFORE
Show person, sensor/input, and visible result.

TRIGGER
Show person, sensor/input, and visible result.

RESPONSE + RESET
Show person, sensor/input, and visible result.

First prototype question
Can ____ cause ____ reliably? Missing test / owner / next school-time check: ____.

First build roles	Names / contribution
Hardware guardian	
Coder / logic explainer	
Test and evidence lead	

Mentor conversation - not a score
Record Approved / Revise and Return / Needs Technical Review. Name the next action; untested hardware remains pending.